SE LAB-01 Problem Statement #19

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Section 1:

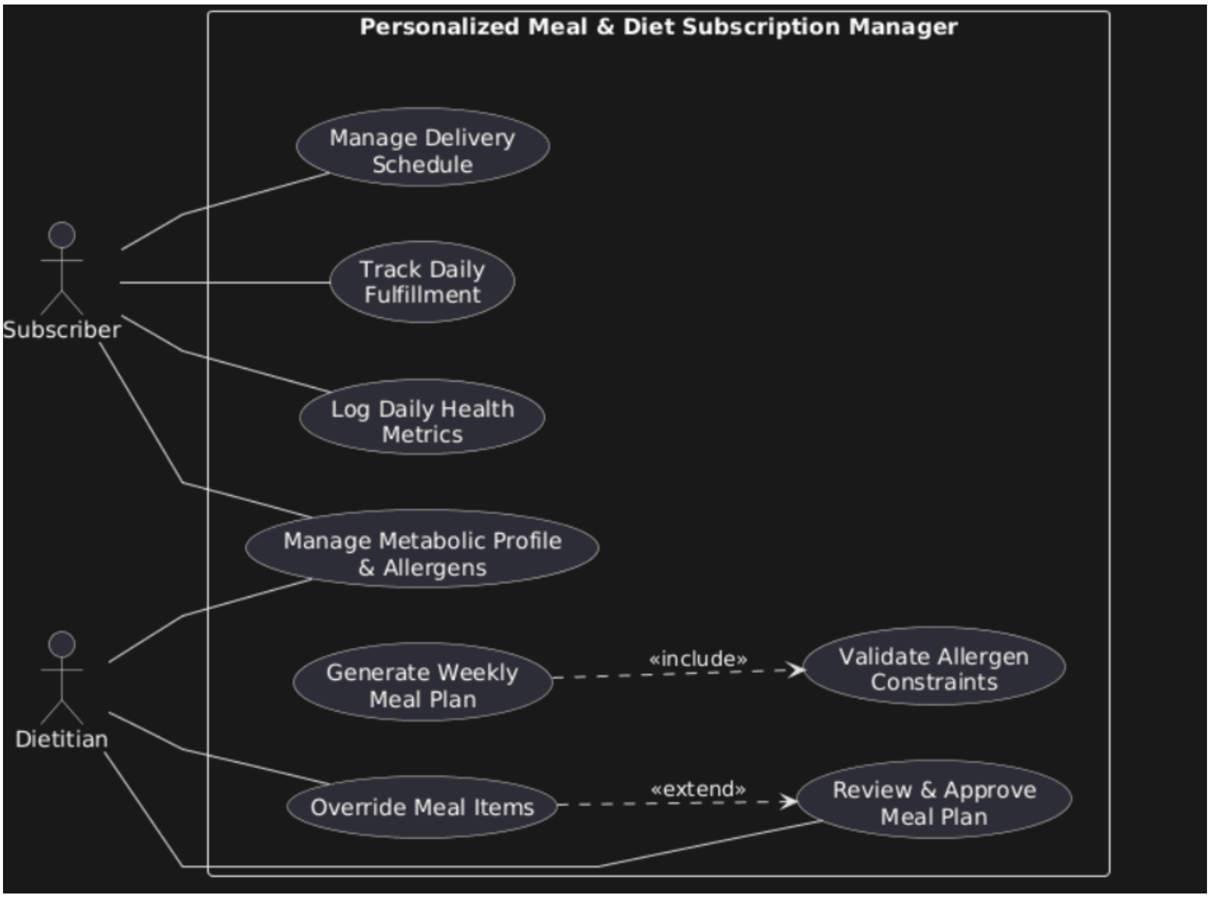
Requirements Table

ID	*Type*	*Description*	*Priority*	*Acceptance Criteria*	*Rationale*
FR-001	Functional	The system shall: generate a personalized weekly meal plan based on the subscriber's metabolic profile, calorie goal, and allergen exclusions.	High	When a plan is generated, it contains no excluded allergens and stays within ±5% of the subscriber's calorie target.	Ensures dietary safety and calorie compliance.
FR-002	Functional	The system shall: allow a dietitian to review, modify, and approve a meal plan before dispatch scheduling.	High	When the dietitian selects “Approve Plan,” the plan status changes to “Approved” and the dietitian ID and timestamp are recorded.	Provides required clinical oversight.
FR-003	Functional	The system shall: allow a subscriber to log daily blood glucose, weight, and blood pressure measurements.	Medium	When valid measurements are submitted, they are saved to the subscriber profile and displayed on the clinical dashboard.	Provides data for ongoing dietary recommendations.
FR-004	Functional	The system shall: allow a subscriber to pause, skip, or reschedule a delivery more than 12 hours before dispatch.	High	When a schedule change is made more than 12 hours before dispatch, the change is accepted and the delivery is marked “Skipped” or “Paused.”	Reduces food waste and supports schedule changes.

FR-005	Functional	The system shall: track daily delivery status and notify the subscriber when the status changes.	Medium	When a delivery changes to “Prepared,” “Dispatched,” or “Delivered,” the new status and timestamp are recorded and a notification is sent.	Keeps subscribers informed about delivery progress.
NFR-001	Performance	The system shall: generate a weekly meal plan within 2.5 seconds at peak load with 500 concurrent requests.	High	A load test shows that the 95th-percentile generation time is ≤ 2.5 seconds with 500 concurrent requests.	Ensures responsive meal-plan generation.
NFR-002	Security & Compliance	The system shall: protect health records, metabolic constraints, and dietary histories using AES-256 at rest, TLS 1.3 in transit, and RBAC.	High	A security test verifies that health data is encrypted and inaccessible to unauthenticated or unauthorized users.	Protects sensitive clinical information.

Section 2:

UML Diagram



Section 3:

Use case diagram

Use Case ID: UC-04

Use Case Name: Review & Approve Meal Plan

Primary Actor: Dietitian

Secondary Actor: Subscriber (Recipient)

Preconditions:

1. The Dietitian is authenticated into the clinical dashboard.
2. The automated engine has generated a draft weekly meal plan (UC2) with validated allergen constraints (UC3).
3. The Subscriber's profile and metabolic constraints are active in the system.

Postconditions:

1. Meal plan status is set to "Approved".
2. Kitchen preparation and delivery schedules are locked in for the upcoming cycle.
3. The Subscriber receives an approval notification and can view the finalized menu.

Main Success Scenario:

1. Dietitian accesses the pending review queue from the clinical dashboard.

2. System displays the list of pending subscriber meal plans.
3. Dietitian selects a specific subscriber's weekly plan.
4. System displays the daily meal breakdown alongside the subscriber's metabolic parameters and allergen constraints.
5. Dietitian reviews macro-nutrient breakdowns, portion limits, and diabetic thresholds.
6. Dietitian selects "Approve Plan".
7. System validates that all mandatory clinical checks pass, saves the "Approved" state, and logs the dietitian's ID and timestamp.
8. System sends a notification to the Subscriber confirming the meal schedule.
9. Use case ends successfully.

Alternate Flow:

5a. Meal Plan Modification / Override

5a1. Dietitian determines that a meal item does not meet the subscriber's specific clinical goals, such as sodium level being too high.

5a2. Dietitian selects "Override Meal Items".

5a3. System displays eligible replacement items filtered according to the subscriber's allergen exclusion rules.

5a4. Dietitian selects a replacement item and enters a clinical note.

5a5. System recalculates the total weekly macro-nutrient values in real time.

5a6. Dietitian reviews the updated meal plan.

5a7. Dietitian resumes at Step 6 of the Main Success Scenario.